

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		 or equivalent (1)	1			1		
		(ii)		T A G C A T One mark for both A-T One mark for C-G		2		2		
		(iii)	I	Guanine 16.7 - 17.7% Cytosine 16.7 - 17.7% Thymine 32.3 – 33.3%		1		1	1	
			II	Complementary (base pairing) (1) (For all organisms,) G is very close in {composition/percentage} to C and A is close in composition to T (1) Variation from exact match due to {experimental error/ free nucleotides in cell} (1)		2	1	3		1
	(b)			Each strand acts as a template (for a new strand) (1) (Newly synthesised) DNA (molecule) contains one old and new strand (1)	2			2		

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	(c)			Any four (×1) from A. {Triplet code/ 3 bases code} for one amino acid (1) B. Explanation of why a minimum of three bases is needed (1) C. There are {64/ many} combinations of the 4 bases/ (genetic code is) degenerate/ or description of (1) D. Different {base / codon} sequences will give rise to different amino acid sequences (1) E. {DNA sequence/ genes} can be varying lengths (1) F. Different amino acid sequences lead to a wide variety of different {proteins/ polypeptides} / ref to exons or introns (1)	2	2		4		
				Question 5 total	5	7	1	13	1	1